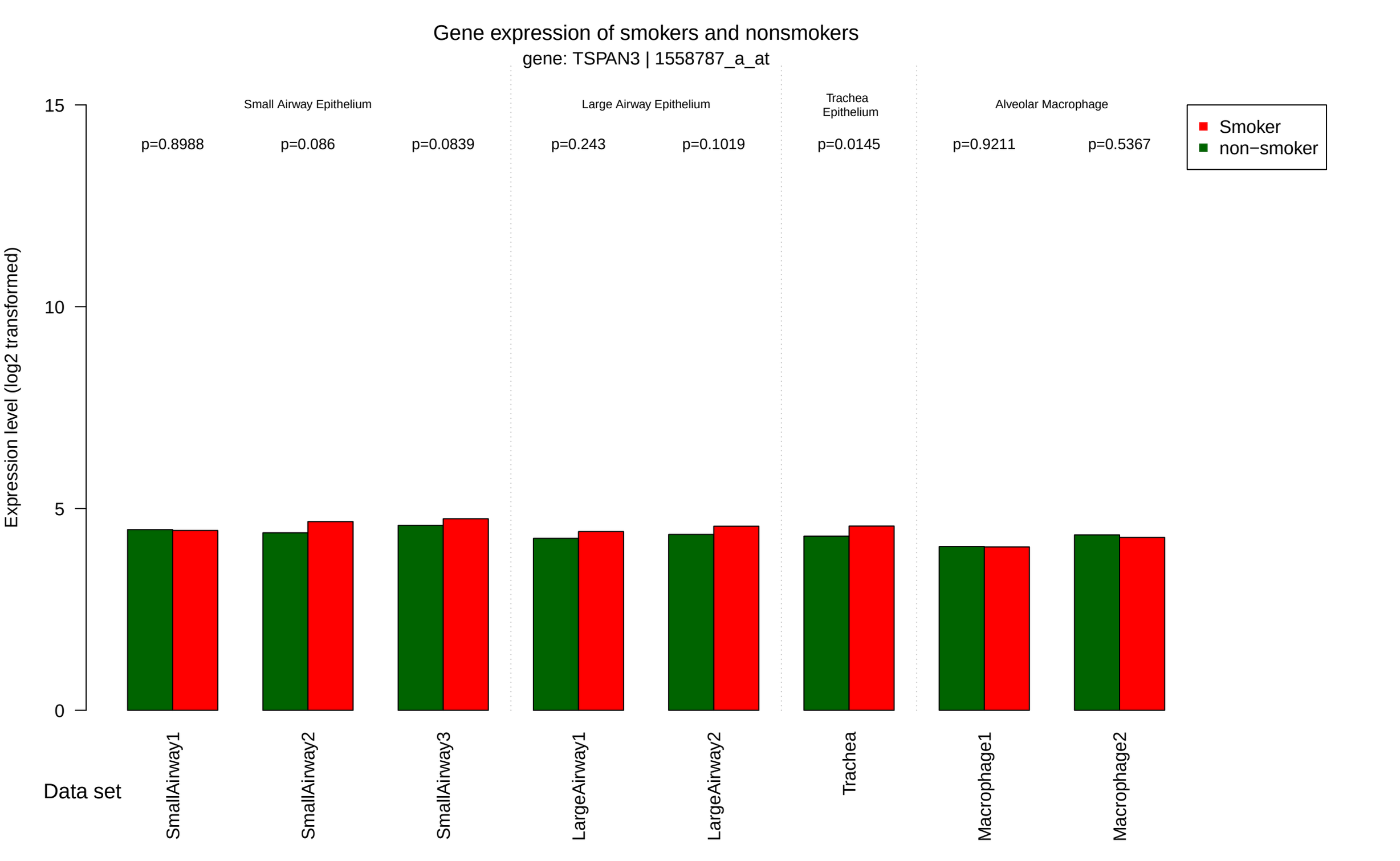
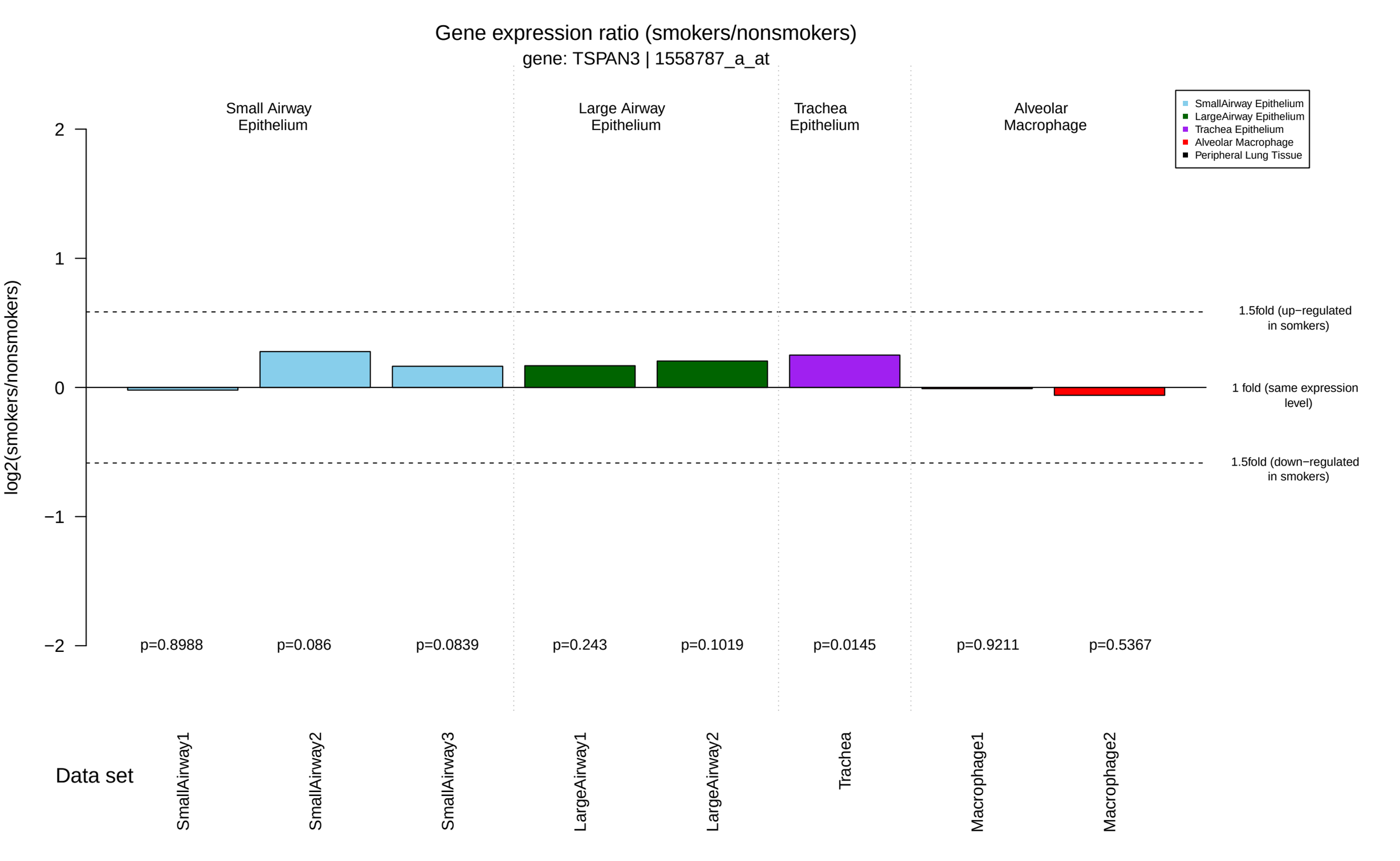
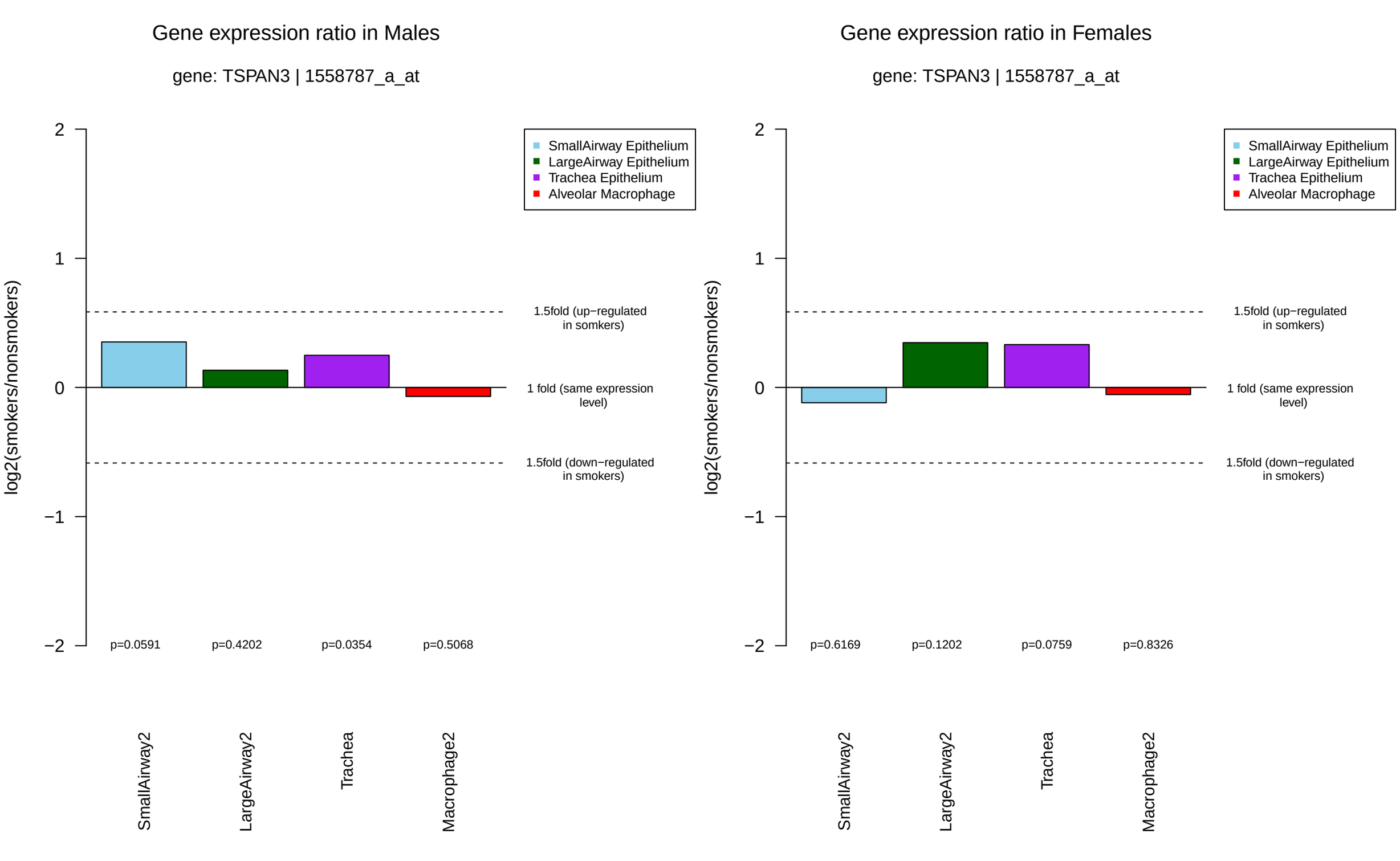
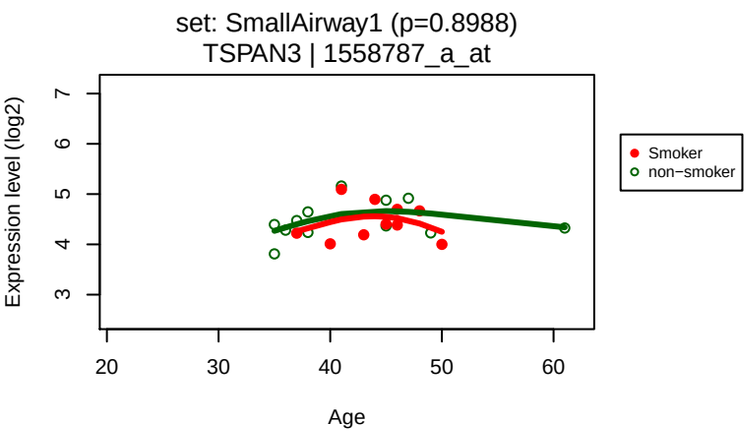
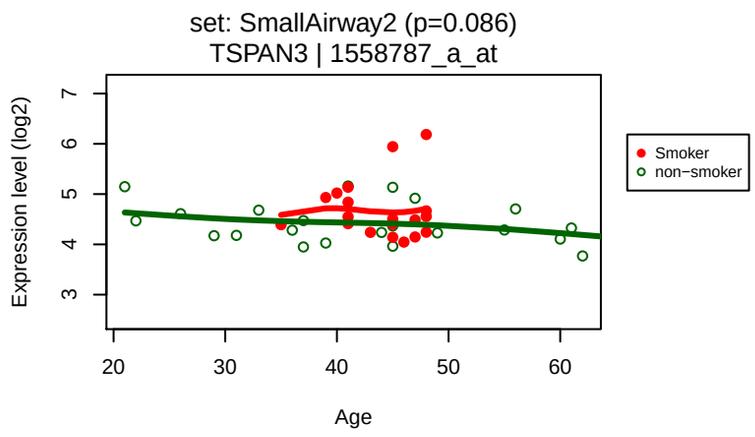
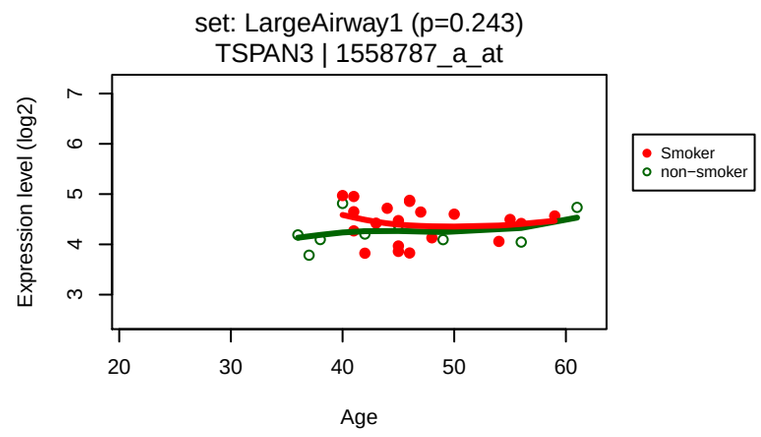

Gene expression of smokers and nonsmokers
gene: TSPAN3 | 1558787_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8988
p=0.086
p=0.0839
p=0.243
p=0.1019
p=0.0145
p=0.9211
p=0.5367
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

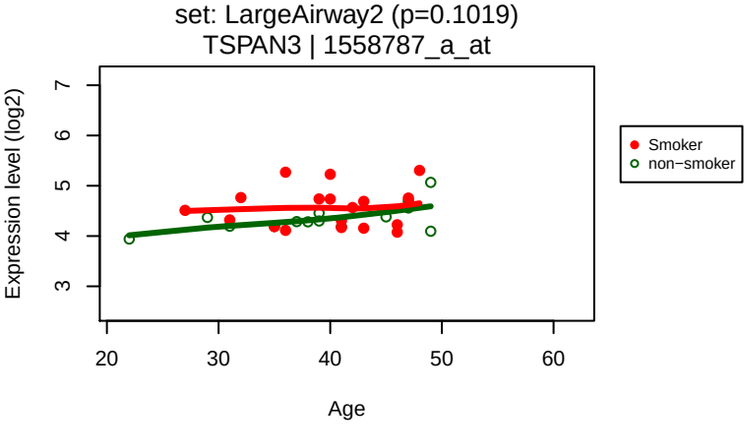

Gene expression ratio (smokers/nonsmokers)
gene: TSPAN3 | 1558787_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8988
p=0.086
p=0.0839
p=0.243
p=0.1019
p=0.0145
p=0.9211
p=0.5367
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

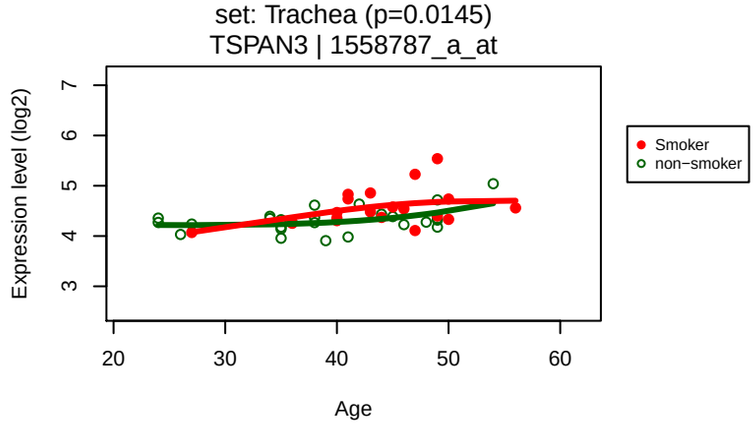

Gene expression ratio in Males
gene: TSPAN3 | 1558787_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0591
p=0.4202
p=0.0354
p=0.5068
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TSPAN3 | 1558787_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6169
p=0.1202
p=0.0759
p=0.8326
SmallAirway2
LargeAirway2
Trachea
Macrophage2

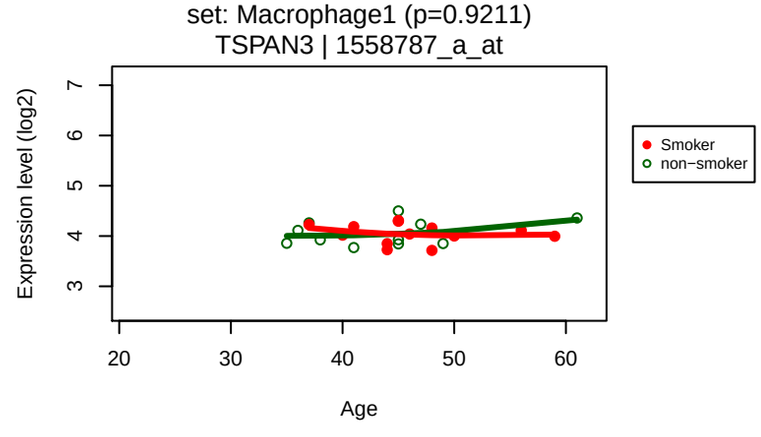

set: SmallAirway1 (p=0.8988)
TSPAN3 | 1558787_a_at
Expression level (log2)
Age
Smoker
non−smoker

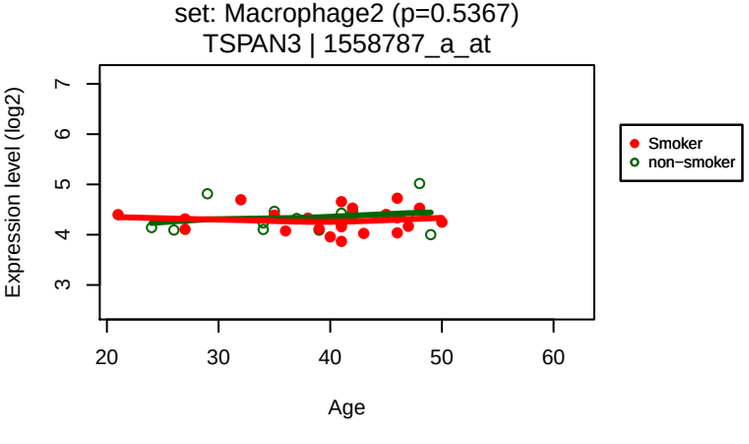

set: SmallAirway2 (p=0.086)
TSPAN3 | 1558787_a_at
Expression level (log2)
Age
Smoker
non−smoker

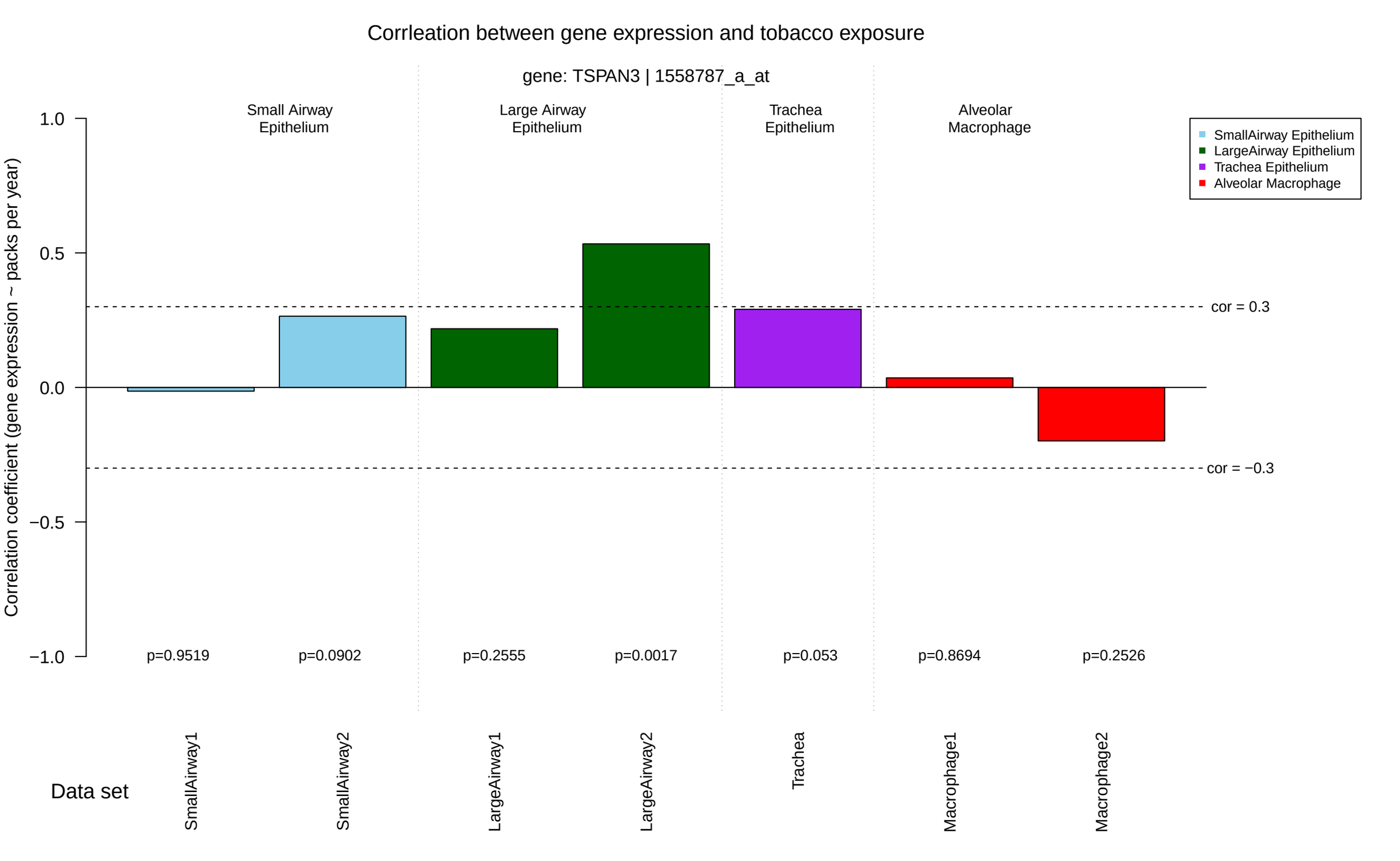

set: LargeAirway1 (p=0.243)
TSPAN3 | 1558787_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1019)
TSPAN3 | 1558787_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0145)
TSPAN3 | 1558787_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9211)
TSPAN3 | 1558787_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5367)
TSPAN3 | 1558787_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TSPAN3 | 1558787_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9519
p=0.0902
p=0.2555
p=0.0017
p=0.053
p=0.8694
p=0.2526
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2